MY EQUATION IMPACT REPORT

RESIDENT ROBOTICS PROGRAM

The Resident Robotics Program

Cumulative Impact Report

Three cohorts. One continuous program. One growing community of robotics builders.



The Resident Robotics Program team and cohort together

My Equation : Tech Analogy Pvt. Ltd. : Ahmedabad

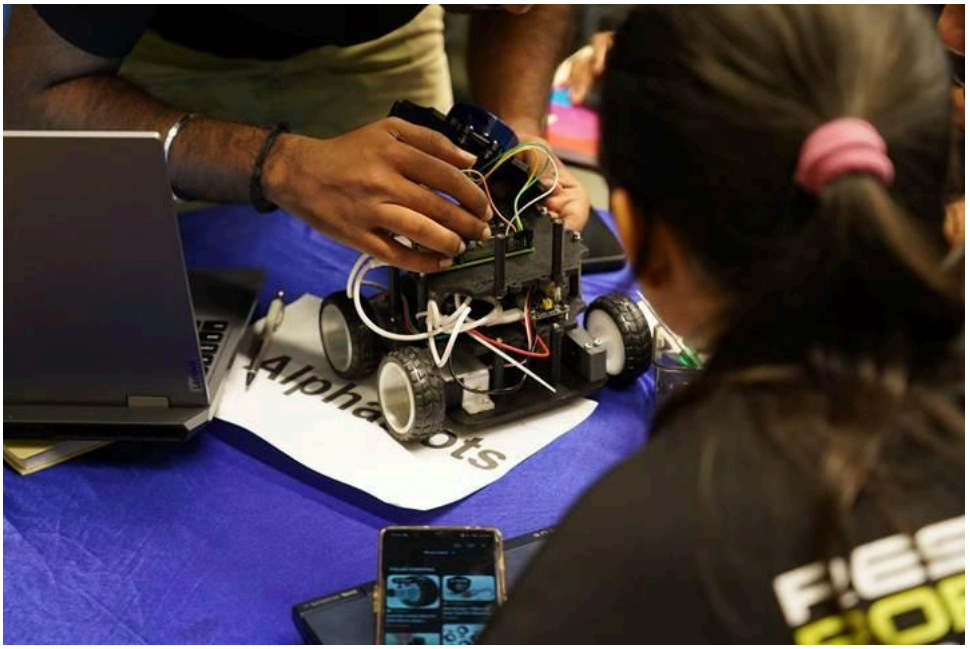
02 IMPACT AT A GLANCE

Three Cohorts, One Growing Impact

Since its pilot cohort, the Resident Robotics Program has run as a single continuous initiative, not three separate courses. Each cohort has built on the last: the same immersive residency model, a growing bench of expert mentors, and an expanding set of flagship robotic systems, delivered to a steadily widening pool of learners across Ahmedabad.

Across all three cohorts to date, the program has trained 195 participants, engaged 10 expert mentors and judges, delivered 47 days of immersive, hands-on training, and conducted 3 industry visits, all in service of one mission: turning learners into engineers who can take a robot from concept to working system.





Participants assembling a robot chassis during the hands-on build phase

03 LEARNING MODEL

The RRP Learning Model

Every cohort of the Resident Robotics Program follows the same underlying arc: participants move from foundational concepts to a functioning robot, one deliberate stage at a time. This consistency is what allows each new cohort to build on the lessons of the last.



How the Model Works

Foundations to Intelligence. Participants begin with the mechanical and mathematical principles behind every robot, then move into the electronics and perception that let a robot sense its surroundings, and the software, AI, and ROS architecture that let it think and act.

Build, Don't Just Learn. The program is structured around building. Each cohort spends its hands-on phase assembling, integrating, testing, and iterating on a real robotic system in a team, not a simulation exercise.

Career Readiness Built In. Every cohort includes dedicated career development: resume building, LinkedIn presence, and professional communication, so technical growth translates into professional readiness.

Real Industry Contact. Each cohort has included a visit to a working industry facility, connecting classroom and lab work to how robotics and automation actually operate at scale.

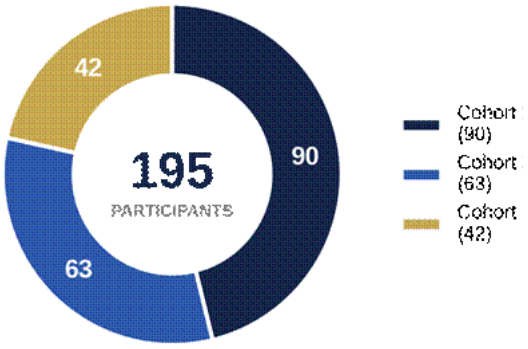


A mentor introducing the Resident Robotics Program's hands-on approach

04 STUDENT JOURNEY

Student Learning Journey

The program has consistently drawn a wide mix of learners, from school students and 12th-pass learners to undergraduates across every year of study, postgraduates, and working professionals. Every cohort has spanned academic and technical streams including Mechanical Engineering, Electronics, Computer Science, AI and Machine Learning, Robotics and Automation, Data Science, and Embedded Systems.



The program's reach has expanded well beyond its 42-participant pilot: Cohort 2 nearly doubled that scale, and Cohort 3 sustained a large, diverse intake while broadening participation among working professionals.

Across all three cohorts, the mix of first-time learners and experienced professionals in the same room has consistently mirrored how real engineering teams are built, and pushed every participant to learn from peers with different starting points.

A Consistent, Multidisciplinary Mix

- School students and 12th-pass learners exploring robotics for the first time.
- Undergraduate engineering students across every year of study.
- Postgraduates and working professionals upskilling alongside their careers.



A mentor-led classroom session during the technical learning phase

05 TECHNICAL LEARNING

Technical & Hands-on Learning

Across every cohort, the curriculum has covered the same three connected layers that make a robot work, taught in the order a real engineering team would need them.

Mechanical & Design

Robot structures, kinematics, CAD in tools such as Onshape and Fusion, material selection, and manufacturing methods, the physical layer of every build.

Electronics & Perception

Embedded platforms including ESP32, Raspberry Pi, and Jetson; sensors, actuators, and IoT integration; and computer vision with OpenCV and YOLO, the sensing layer that lets a robot understand its surroundings.

Software, AI & Autonomy

ROS 2 fundamentals, simulation in Gazebo, SLAM and navigation, motion planning with MoveIt2, and AI integration, the intelligence layer that lets a robot plan, move, and act on its own.

> **Every cohort has ended its technical phase the same way: with a team applying the full stack, mechanical, electronics, and software, together on one working robot.**



Hands-on assembly and integration



Debugging and wiring a mobile robot platform

06 PROJECTS & APPLICATIONS

Projects & Applications

Across three cohorts, participant teams have designed, built, and demonstrated four flagship robotic systems, each one a real engineering challenge, not a simulation.

Robotic Arm Manipulator

Built in every cohort. A robotic arm capable of controlled, precise motion and pick-and-place operations, with every joint calibrated for accurate, repeatable movement. One Cohort 1 team's implementation won the program's top project award.

TurtleBot / T-Bot

A mobile robot platform used to implement and validate the fundamentals of autonomous mobility, navigation, and system bring-up on a standard robotics base.

Campus Delivery Robot

A mobile robot designed to navigate a real environment and move reliably from point to point without human control, combining SLAM, mapping, and autonomous navigation.

Blinking & Moving Eye Robot

A perception and expression build: an animated robotic eye that blinks and moves in real time in response to a live camera feed, the program's most creative computer vision application.



A project team with their completed build

07 INDUSTRY EXPOSURE

Industry & Expert Exposure

Throughout its history, the program has connected classroom learning to real industry practice, through both expert-led sessions and on-site industry visits.

Industry Visits

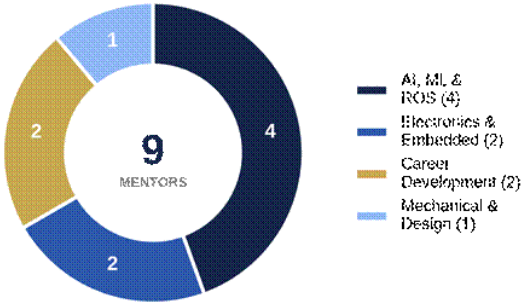
- **Robologic, Ahmedabad.** Visited early in the program for first-hand exposure to industrial robotics applications, automation systems, and manufacturing workflows.
- **Voltas Beko.** Visited on multiple occasions for exposure to industrial manufacturing and automation at scale.



Participants on an industry visit, seeing manufacturing and automation at scale

Expert Mentors & Judges

Across the program's history, 9 expert mentors have led sessions and 1 industry judge has evaluated final demonstrations, spanning mechanical design, electronics, AI and machine learning, ROS, and career development, including practitioners from Honeywell, NIT Trichy, and ALLBOTIX.



08 SKILLS & DEVELOPMENT

Skills & Development Framework

Every cohort has been evaluated against the same two-sided framework: the technical capability to build a working system, and the professional readiness to present it.

Technical Competencies

- Robotics engineering fundamentals
- Mechanical design and CAD
- Electronics and embedded integration
- Computer vision and AI
- ROS 2 and autonomous navigation
- Robotic manipulation

Professional Competencies

- Professional communication
- Resume and LinkedIn presence
- Personal branding
- Industry and workflow awareness
- Team collaboration and ownership
- Professional confidence

How Progress Was Measured

Every cohort was evaluated continuously through its hands-on phase, not just on a single demo, against five consistent dimensions:

- Technical execution
- Innovation
- Problem-solving
- Teamwork
- Presentation quality

09 DATA INSIGHTS

Data Insights

The program has deliberately expanded the depth of its curriculum over time, growing from two flagship robotic systems to a sustained set of four, without diluting the rigour of any single track.

> **The Robotic Arm Manipulator track has run throughout the program's history and produced its highest-recognised project to date, a team implementation that won the program's top project award.**

A Consistent Evaluation Framework

Every team, on every project, has been assessed against the same five dimensions, ensuring recognition reflects sustained engineering effort rather than a single demonstration.



The program has also steadily grown the pool of people directly invested in participant outcomes, expanding its mentor bench and, most recently, adding a dedicated external judge to its final evaluation panel.

10 OVERALL IMPACT

Overall Impact & Closing Reflection

Across three cohorts, the Resident Robotics Program has taken 195 participants through the same journey: exposure to a new discipline, structured learning, hands-on building, real experimentation, applied engineering, public demonstration, and, for many, a lasting shift in how they see their own capability.

The program has not stood still. Each cohort has widened its reach, deepened its curriculum, and strengthened its bench of mentors and industry partners, while holding on to the same core belief: the fastest way to learn robotics is to build one.



Final project evaluation on Demo Day

One mission: to build the next generation of robotics engineers.